

**Proposal for genetic characterisation
of wild-type mumps strains: Preliminary
standardisation of the nomenclature**

Brief Report

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Received May 28, 2004; accepted April 14, 2005

Published online June 15, 2005 © Springer-Verlag 2005

Summary. Though mumps virus (MuV) is a monotypic virus, genetic variation between strains has been described. Viruses have been placed into genotypes designated A–L based on the nucleotide sequence of the small hydrophobic (SH) gene, which is the most variable gene in the mumps genome. Molecular characterisation of MuV is an important component of mumps surveillance because it can help identify the transmission pathways of the virus as well as distinguish between wild-type and vaccine strains. Here, we propose a standardized nomenclature and an analysis protocol for the genetic characterisation of mumps strains to facilitate expansion of molecular epidemiological studies. In addition to assigning standard

reference strains for the recognized genotypes of MuV, a convention is proposed for naming for strains and criteria to designate a new genotype.

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Mumps virus (MuV) is a member of the genus *Rubulavirus* of the family *Paramyxoviridae*. Its genome is a single-stranded, negative sense, non-segmented RNA of 15,384 nucleotides. The genome contains seven transcription units that encode open reading frames (ORFs) for the nucleocapsid (N), the V protein (and after editing the phospho- (P) protein), the matrix (M), the large (L), the fusion (F), the haemagglutinin-neuraminidase (HN) and the small hydrophobic (SH) proteins [3].

Mumps is a common and highly transmissible viral disease with infection resulting in bilateral parotitis. Infections are not usually fatal, but the neurological complications of mumps can be severe. Aseptic meningitis occurs in about 10% of cases. In 0.02–0.03% of cases, encephalitis may occur and require hospitalization. In fact, mumps encephalitis accounted for 36% of viral encephalitis cases before the introduction of vaccine. In the pre vaccine era, annual incidence rates for mumps were 200–700/100,000 and rates of up to 6000/100,000 have been reported in military and community outbreaks. In countries that have successfully introduced the MMR vaccine or monovalent mumps vaccine, incidence rates for mumps are usually less than 1/100,000. MMR is currently used in 23/25 developed countries and 19/22 of countries with economies in transition. However, MMR is only used in 24% (40/168) of developing countries, and very little is known about the burden of mumps disease in countries without mumps vaccination programmes.

Laboratory diagnosis is important for confirmation of mumps infection, as there are other causes of parotitis. The standard methods of serum antibody detection, virus antigen detection in the cells obtained from throat swab and virus isolation from throat swab or urine are still commonly used though in specialised laboratories techniques are now available for the diagnosis of mumps infection by detection of IgM antibody or mumps RNA in oral fluid [8]. Mumps specific PCR products are subsequently sequenced for strain characterisation [2, 7, 8].

MuV is serologically monotypic. Nevertheless, distinct clusters of wild-type viruses have been described and reported to be co-circulating globally. Strain diversity was identified based on a comparison of the nucleotide sequences of the SH gene [23]. The SH gene shows the greatest amount of nucleotide sequence variation on the genome of MuV although the function of the SH protein remains unclear. Phylogenetic comparison of SH gene sequences has been used to classify MuV strains into a number of genotypes. Genotypes A–L (with names proposed in this report) have been described by investigators over recent years [1, 2, 6–13, 15–23].

Investigation of the chronological and geographical distribution of mumps genotypes has shown that MuV genotyping has a valuable role in mumps surveillance [2, 6, 8, 10, 11, 16, 18, 19]. Genetic analysis is the only means available to distinguish vaccine from wild-type viruses. Sequence information is also important for assessing the safety of mumps vaccines since it has been shown that some of the mumps vaccines are more reactogenic than others [4, 5]. The molecular data can also be used to track transmission chains of the virus [8].

Therefore, countries that have introduced mumps vaccination should consider initiating molecular characterization of wild-type mumps viruses as an important component of mumps surveillance.

At present, there is no internationally agreed nomenclature and standard analysis for describing the genetic characteristics of wild-type MuV. Strains have been assigned to various genotypes on the basis of differing sequence windows containing from 270 to 318 nt of the SH gene. Genotypes have been assigned by the investigating laboratory and this lack of standardisation has for example, led two groups, one in Japan [20] and one in Sweden [18] to propose two different strains of MuV as the genotype J at the same time. Scientists from ten laboratories around the world, who have contributed to the genetic characterisation of MuV, propose here a standardization of international nomenclature based on previous reports and experiences of the authors. A system for genetic characterisation of mumps strains that will facilitate future molecular epidemiological studies and avoid confusion in the future designation of new genotypes has also been proposed.

SH genes of several hundred MuV strains have been sequenced. The entire SH gene, 316 nt sequence containing the protein encoding region of 171 nt (57 amino acids) is proposed as the minimum amount of sequence information required to assign a genotype to an unknown strain. Based on a comparison of the published MuV SH sequences available in GenBank and previous proposals, genotypes A–L have been designated. Nucleotide sequence divergence was up to 19% between 30 selected MuV strains, representing genotypes A–L (Fig. 1). The phylogenetic tree (Fig. 1) presents all currently assigned MuV genotypes. The tree was drawn based on 316 nt of the entire SH gene including the non-coding region using the Clustal and MegAlign program of the DNASTAR package (Fig. 1a). Software such as bootstrap analysis (1,000 times) using the Neighbour-joining method (PAUP 4.0 Beta10 package) should be applied in assigning strains to new genotypes (Fig. 1b). Other phylogenetic programmes can also be used for data analysis and genotyping when the reference strains that we propose are employed.

The first isolate from each genotype is proposed to be the reference strain for that genotype. Furthermore, in the phylogenetic trees one recent isolate has also been included when available. The known geographic distributions of these genotypes are listed in Table 1. Some of the 12 chosen reference strains (highlighted in Table 1) may have never been successfully isolated and have been sequenced directly only from clinical specimens. These strains will be replaced in the reference tree by related cell culture isolates if and when they become available. As the Vero cell line is sensitive to mumps virus and available in most laboratories, we suggest that newly proposed reference strains should be isolated in (or adapted to) Vero cells and stored as early passage material.

The strain names should provide basic information as well as be easy to use. The three letters of UN ISO3 country code (<http://unstats.un.org/unsd/methods/m49/m49alpha.htm>), year of detection, ID number of the strain and genotype identified are essential, and MuVs and MuVi indicated at the beginning will represent the strains identified directly in clinical samples or in cell culture isolates respectively, e.g. MuVs-GBR05-320-D or MuVi-CHN06-608-H. Previously published strains can retain their original names as a prefix to the full description.

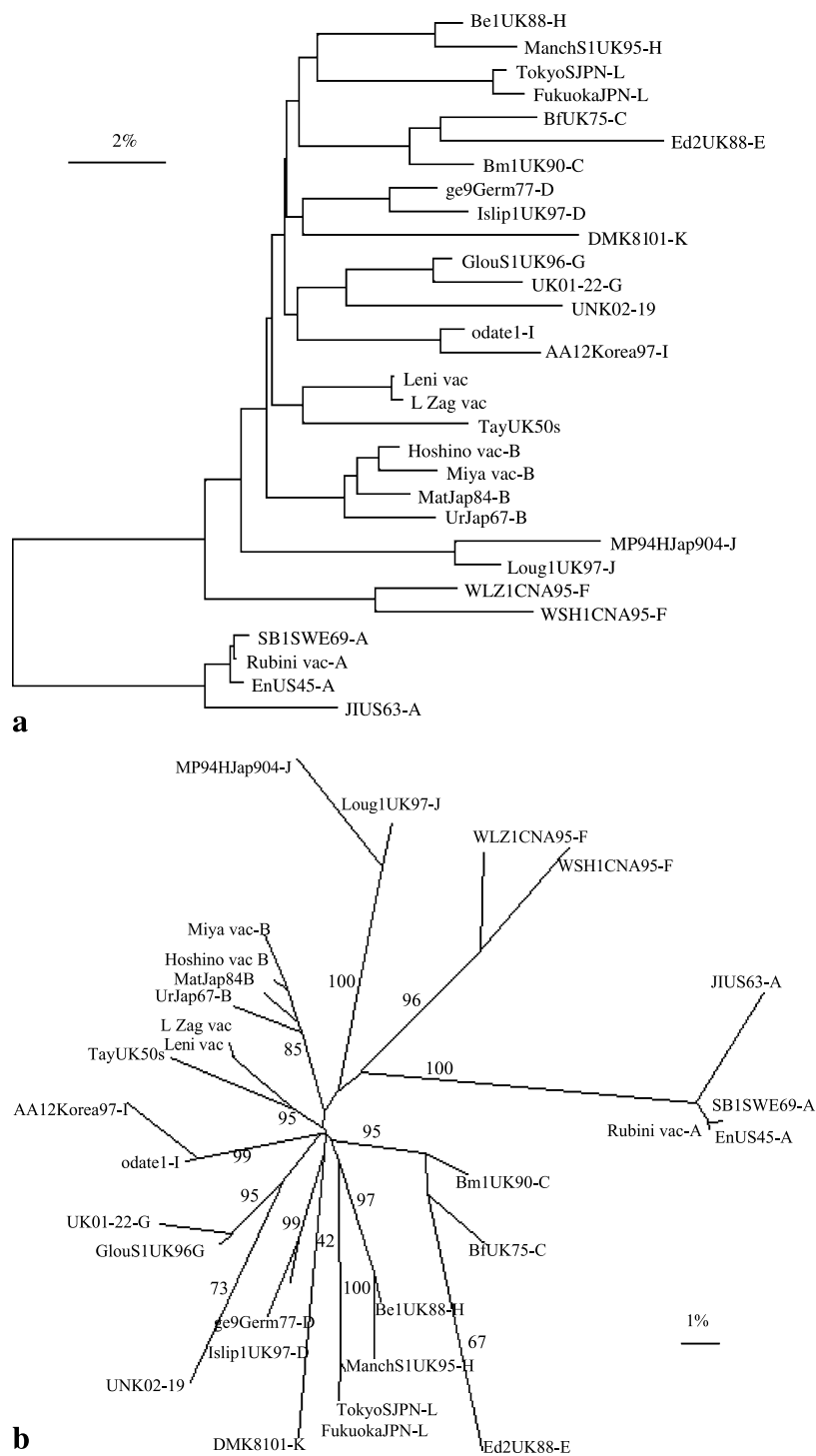


Fig. 1. Phylogenetic tree presenting all current assigned reference strains of mumps virus genotypes including one recent strain when it is available. Genotypes were indicated at the end of each strain. The tree was drawn based on 316 nt of the entire SH gene including the non-coding region (a) using the Clustal and Megalign program of the DNASTAR package and (b) using the bootstrap analysis (1,000 times) of the Neighbour-Joining method (PAUP 4.0 Beta10 package)

Table 1. The proposed reference strains and global distribution of mumps genotypes

Genotype	Reference strain	GB accession no	Country (IS03) and year identified
A	End/USA45* SBL-1/SWE69* JL/US63 (vaccine) Rubini (vaccine)	D90231 D00663 D90232 X72944	USA45, 50, 63; SWE69, 93; CHE74; DEU87, 92; CAN88
B	Urb/Jap67 Mat/Jap84 Miya (vaccine)* Hoshino (vaccine)	D90236 D90233 D90234 AB003414	JPN67-95; GBR89, 90
C	Bf/UK75* Bm1/UK90	X63709 DS26771.DAT	GBR75, 80s, 90, 98-2000; SWE80s, 92; DEU87, 92, 93; CHE95; PRT96; LTU98-00
D	Ge9/Gem77 Islip1/UK97	DS26771.DAT AF142766	DEU77; PRT96; GBR96, 97, 99; LTU99; DEN80s, 90s, 01; JPN93
E (C)	Ed2/UK88	X63711	GBR88
F	WLZ1/CNA95 WSH1/CNA96	Z77158 Z77160	CHN95; GBR 99; SWE71, 72, 84;
G	Glouc1/UK96* UK01-22	AF142764 AY380075	GBR91, 96-05; JPN99-05
H	Be1/UK88 ManchS1/UK95*	DS26771.DAT AF142771	GBR88, 95, 96, 98-01; CHE 95, 98-00; KOR99; JPN97
I	Odate-1 AA12/Korea97*	D86174 AF180374	JPN93; KOR97-01
J	MP94H/JNP94 Loug1/UK97	AB03417 AF142770	JPN94, GBR97
K	DK81/01 (DMK81)	AF365891	DNK81-88
L	Fukuoka49/JPN00* Tokyo S-III-10/JPN01*	AB105483 AB105480	JPN00-01
**	Leningrad 3 (vaccine) L-Zagreb (vaccine)	AY493374 AJ272363	RUS53
**	Tay/UK50s*	AF142774	GBR50s
**	UNK02-19*	AY380077	GBR02

*Isolate available; **Reference strains for potential new genotypes

The proposal presented here was established and finalised through international cooperation of the 13 authors. In future, a new genotype should only be assigned when the following criteria are met:

- (1) The sequence of at least the entire 316 nt of the SH gene is unique when compared to all existing reference strains;
- (2) The divergence (variation rate) based on the 316 nt of the entire SH gene is >5% when compared to all of the established reference strains;
- (3) At least two identical or very similar strains are identified representing the proposed new genotype. New genotypes should be designated if a case can be made for sustained transmission/detection of viruses with significantly divergent SH gene sequences;
- (4) Two other mumps lab/groups must have been contacted for confirmation to avoid duplication of genotype;
- (5) Submit the sequence data to GenBank and indicate the proper strain name and data source (derived from cell culture isolate or directly from clinical sample).

In summary, we proposed the establishment of a standardisation of the nomenclature of MuV strains and the criteria for the assignment of strains to specific genotypes to facilitate molecular epidemiological investigations using these criteria. Genotypes A–L are proposed based on sequence diversity of the SH gene. Sequence analysis of other genes such as the HN or the C terminus of the N protein may also be useful in monitoring MuV evolution and tracking its transmission. Ongoing communication and collaboration between virologists working on MuV are necessary to enhance the role of molecular techniques in surveillance and control of mumps infection. A standardized nomenclature will make it easier for virologist to communicate with public health officials who are involved in mumps control activities.

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